

SOC3050

Atmega128 Setting

Fall 2025

IUT

Hong Jeong

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3. AVR128 Kit Preparation for MC Studio
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5. Data Visualizer Setting
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AVR128 Kit

● 전체 구성 사진

1. MCU 모듈 1EA

2. MAIN BOARD 커넥터 모듈 1EA



1. Jump Cable & Case

2. CDS Module

3. PWM motor controller module

4. Graphic connection module
& Monochrome graphic LCD Module

5. Accelerometer module

6. Axis Variable Resistor Module

7. Proximity sensor module

8. Ultrasonic sensor module

9. Infrared sensor module

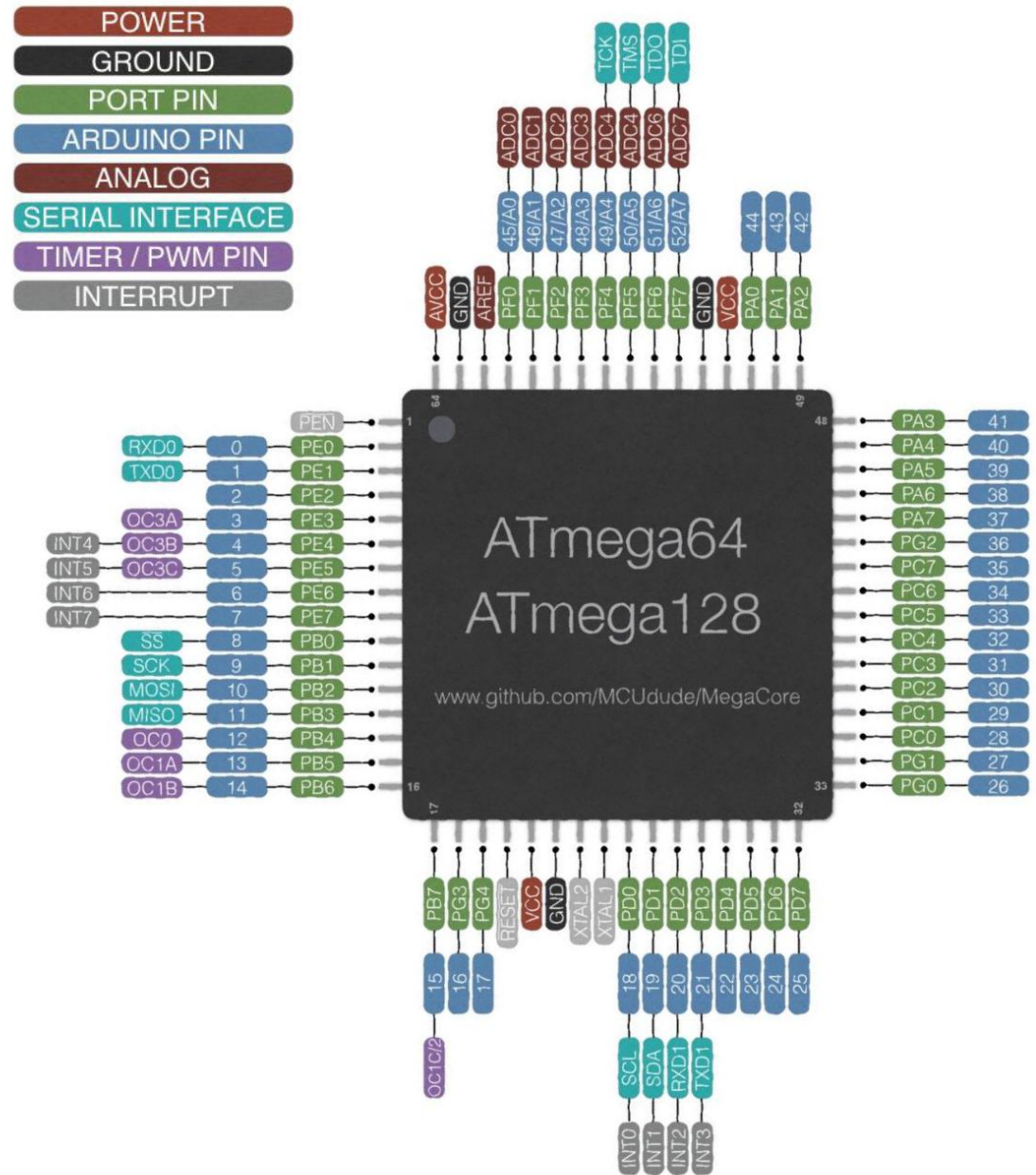
10. Force sensor module

11. Step motor

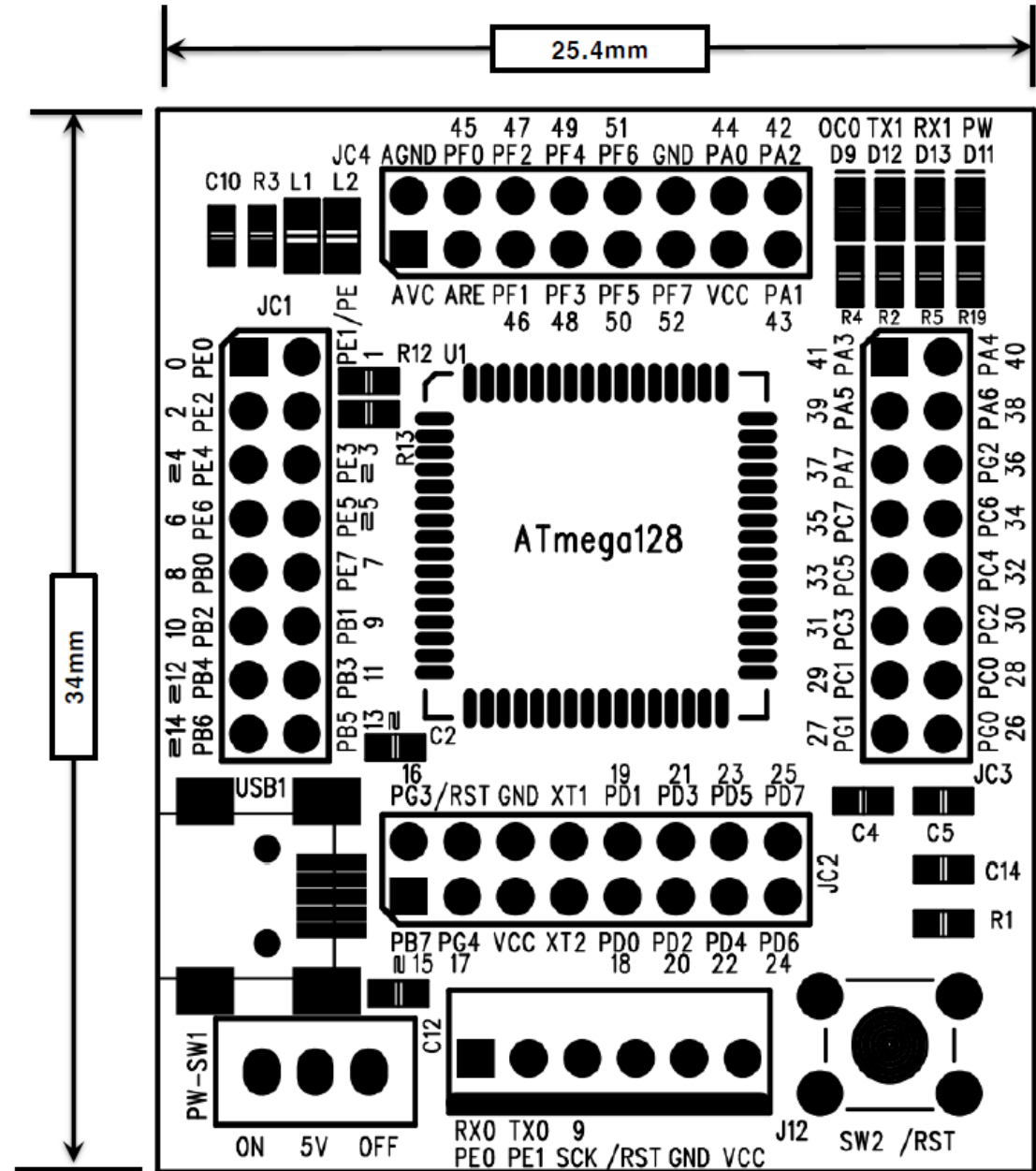
12. DC Motor



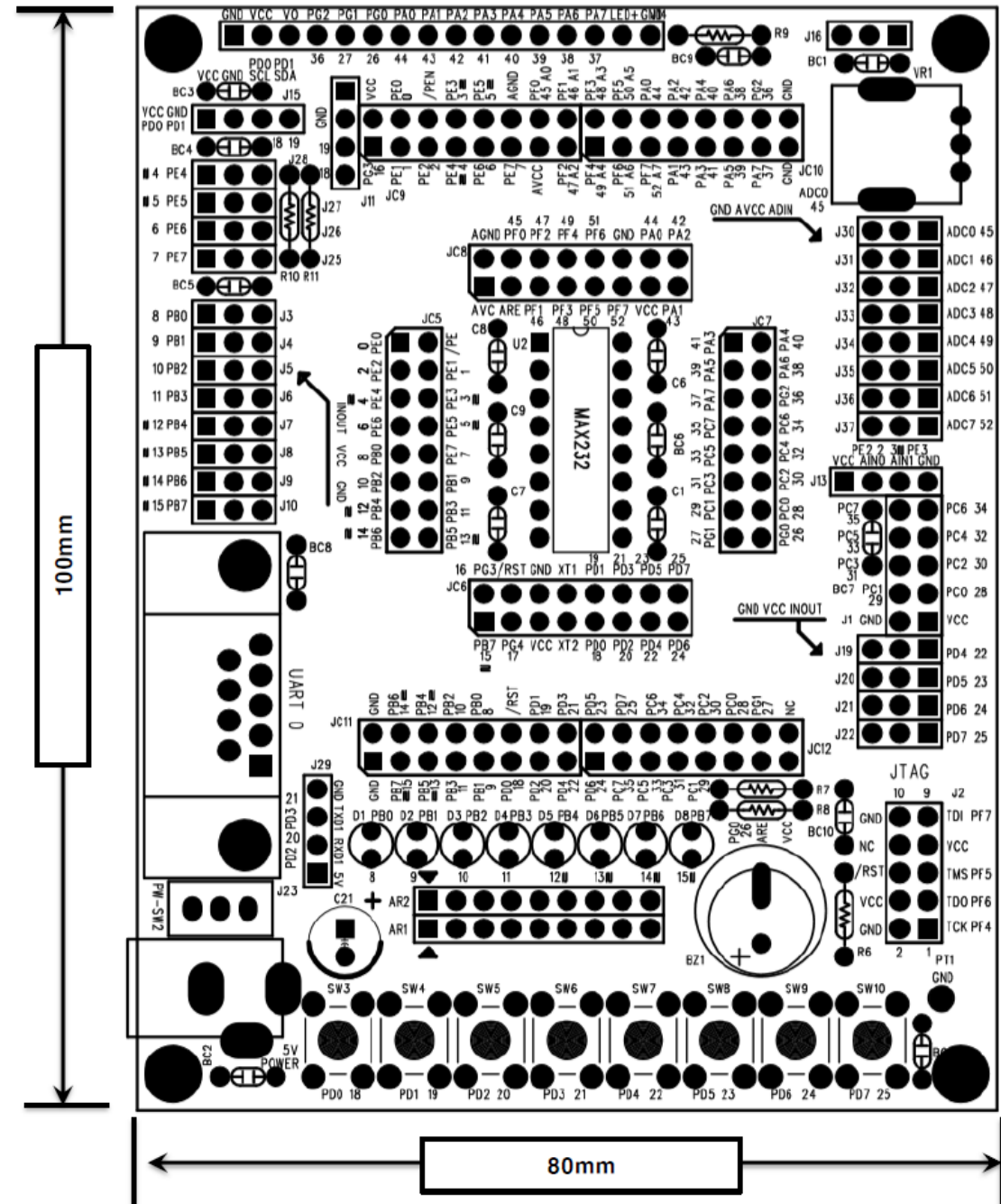
AVR128 MCU

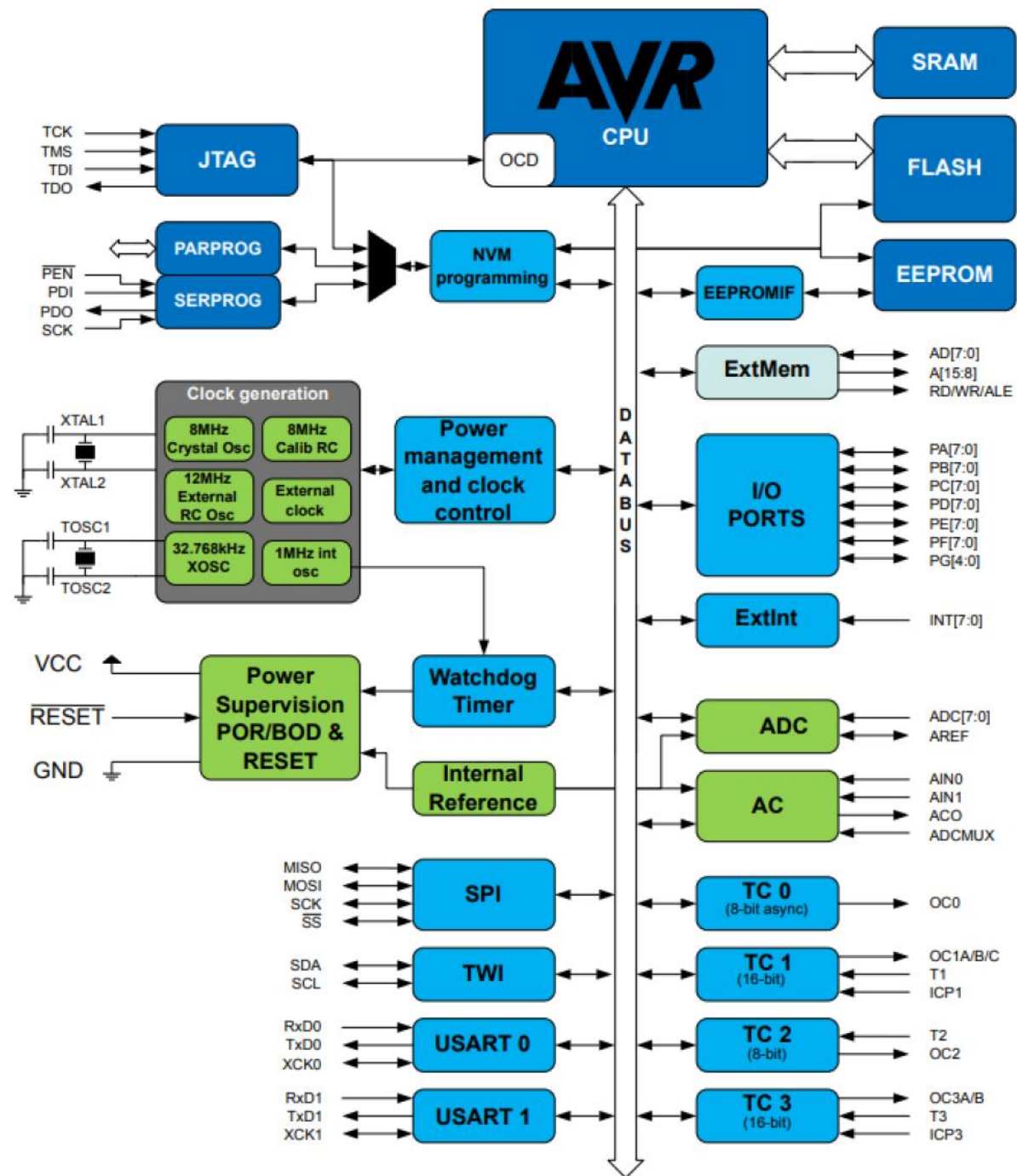


MCU Board



Main Board





Firmware Fix for SimulIDE 1.1.0-SR1

- Fix Error:

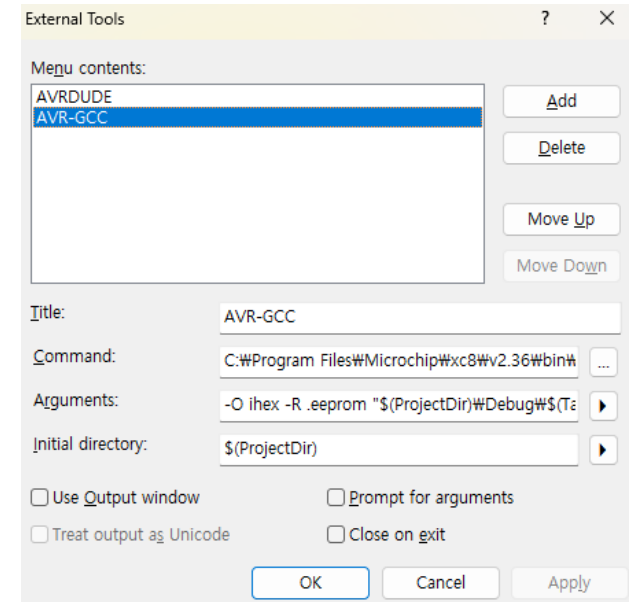
- Warning: Not supported Record type: 2 at line 0

- Command:

C:\Program Files (x86)\Microchip\AVR8 GCC\bin\avr-objcopy.exe

- Arguments:

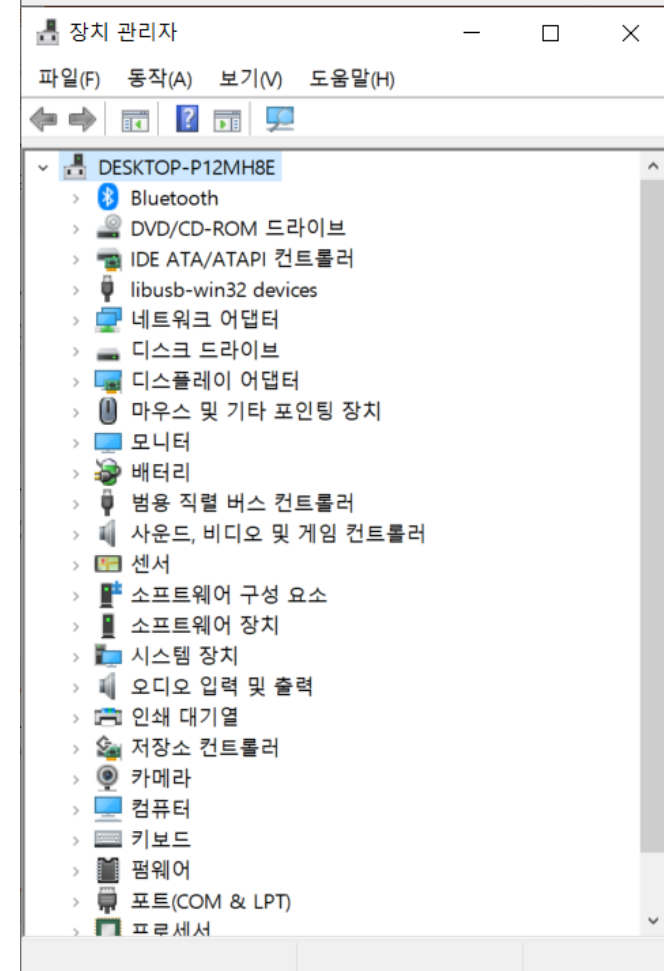
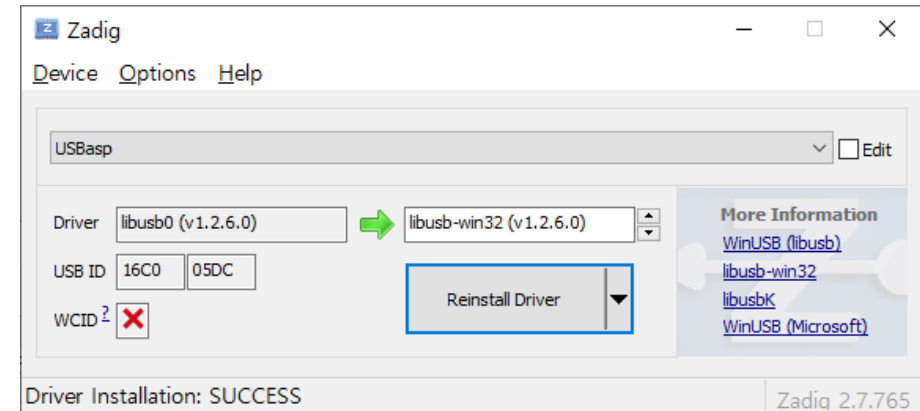
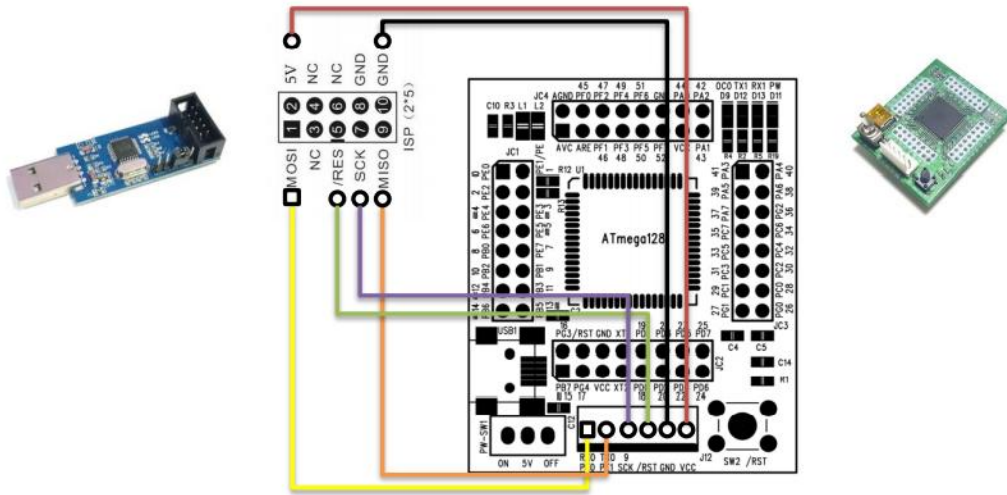
-O ihex -R .eeprom "\$(ProjectDir)\Debug\\$(TargetName).hex"
"\$(ProjectDir)\Debug\\$(TargetName)_clean.hex"



Firmware Uploading

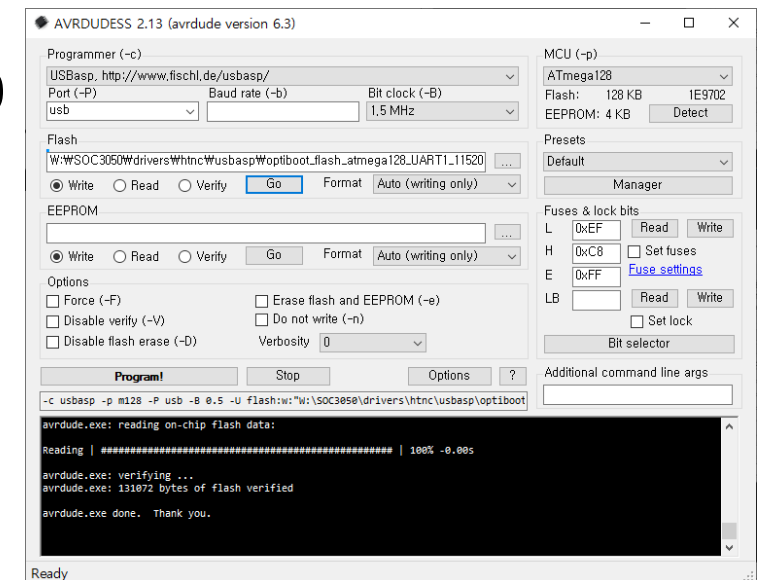
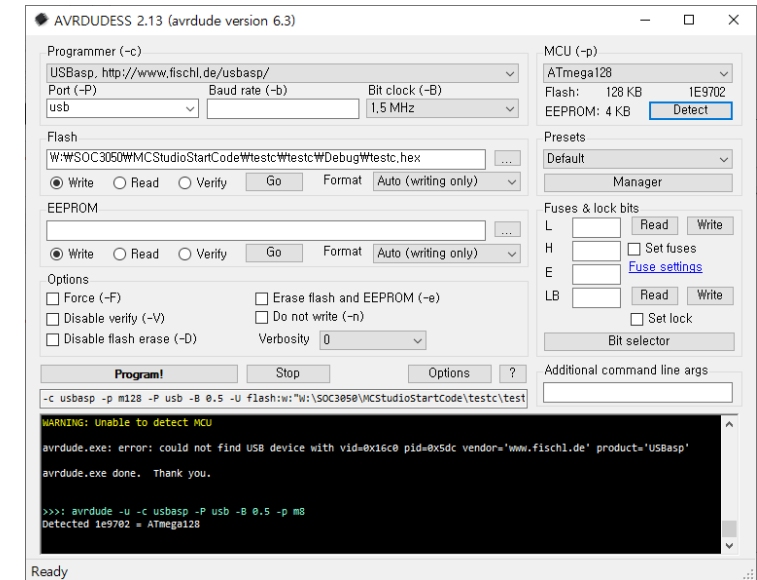
1. USBasp connect

- Plugin USBasp
- Install [Zadig](#) USBasp



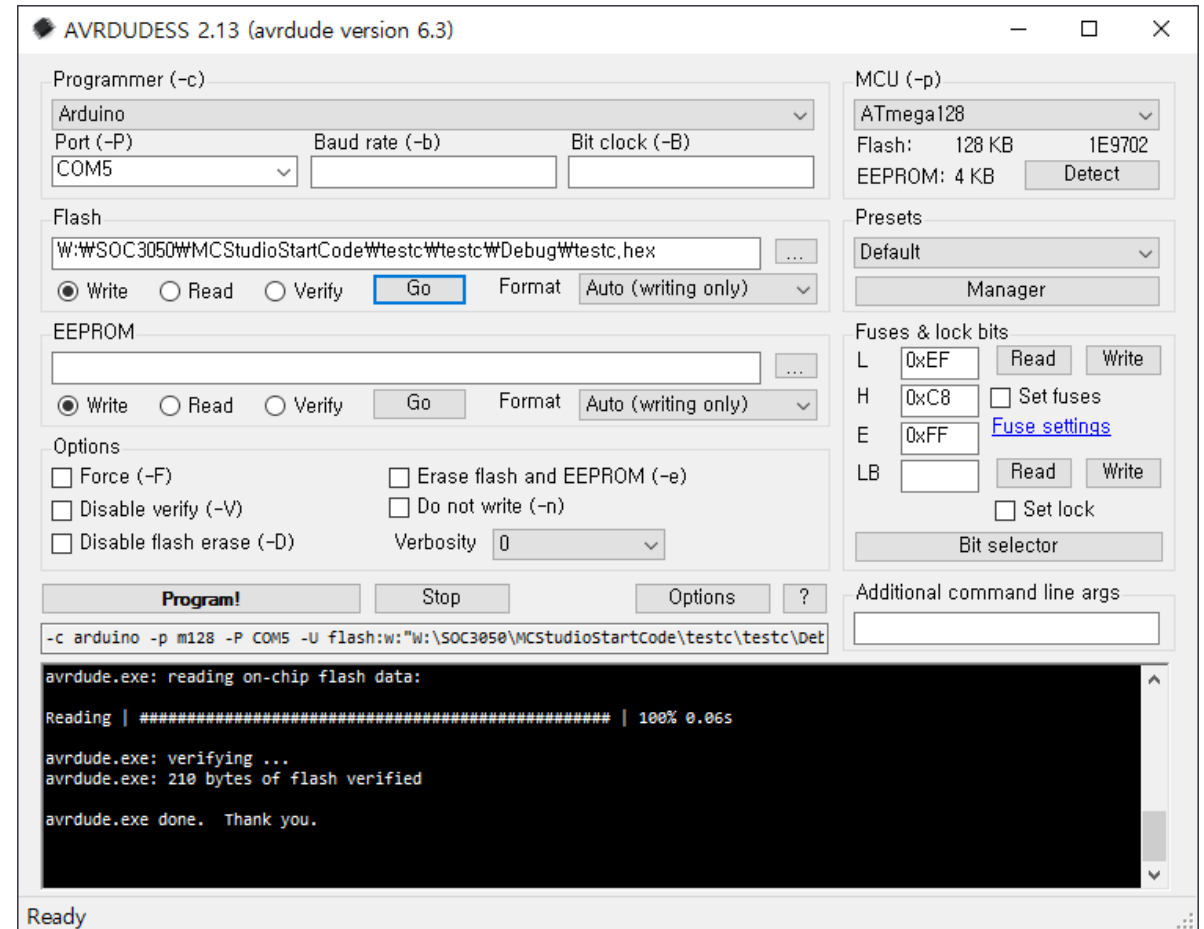
2. Upload Firmware

- Run AVRDUDESS:
- Click Detect
- Click READ
- Write 0xEF, 0xC8, 0xFF
- Upload Firmware
 - Downloa:
<https://github.com/MCUdude/MegaCore>
 - Flash:
optiboot_flash_atmega128_UART1_115200_160000
00L_B5_BIGBOOT.hex
 - Click Go



3. Test Program

- Plugout USBasp
- Plugin Mini USB
- Run AVRDUDESS
 - Detect
 - Programmer: Arduino
 - Port: COMx
 - Flash: blinky.hex
 - Go

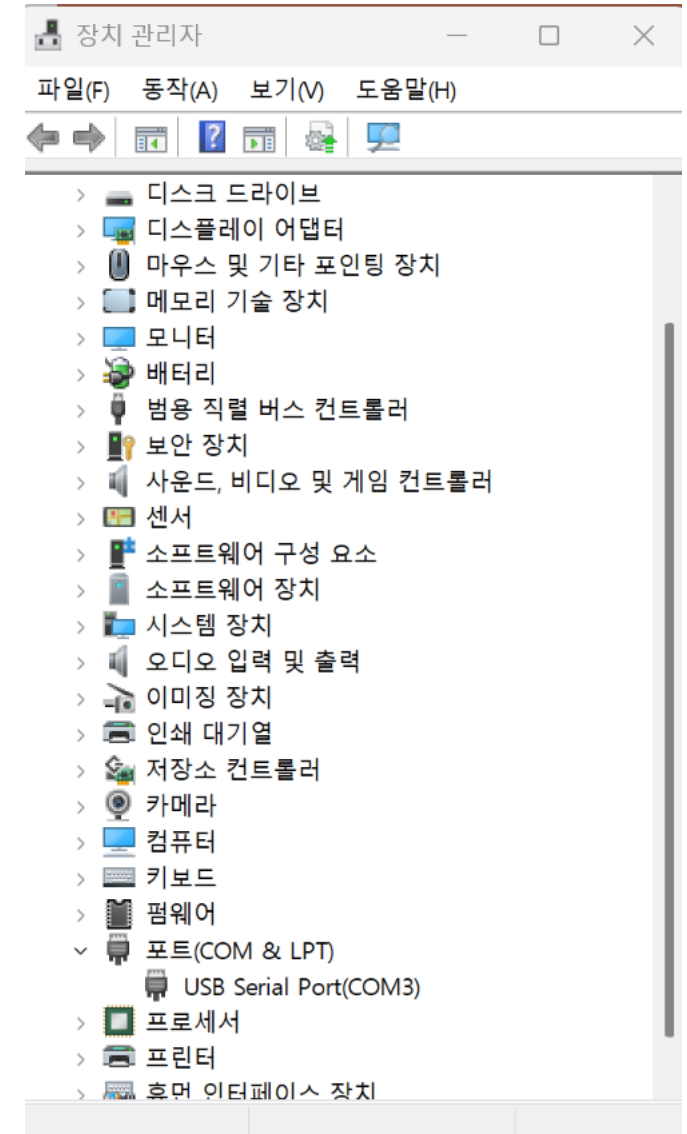
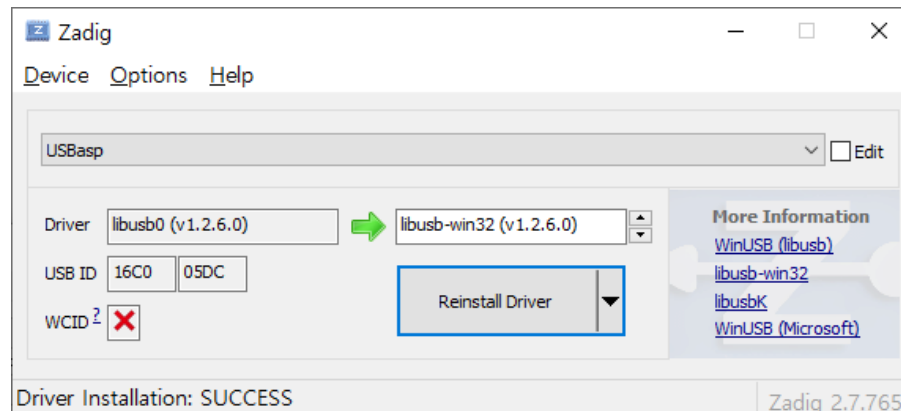


AVR128 Kit Preparation for MC Studio

1. COM Preparation
2. AVRdude connection to COM Port
3. Microchip Studio connection to AVRdude

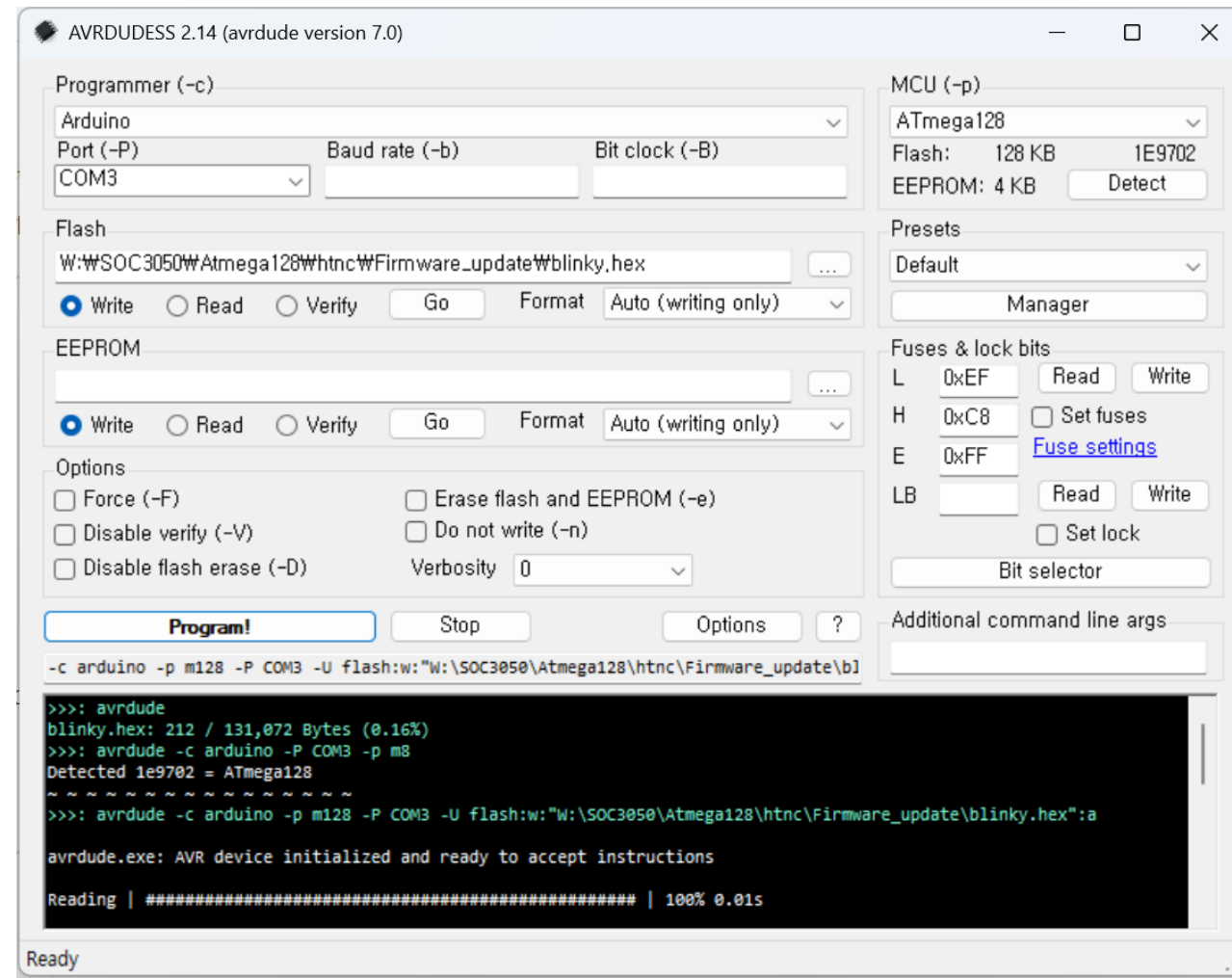
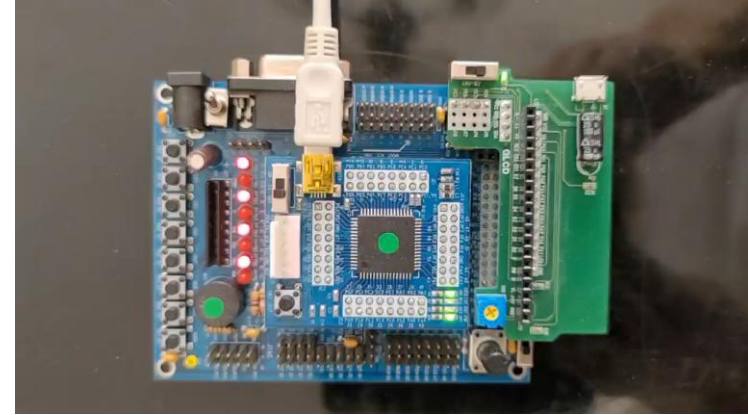
COM Preparation

1. Observe Device Manager for COM Port
2. If COM Port exists, Skip the following stage.
3. Install [Zadig](#) USBasp



AVRDUDESS

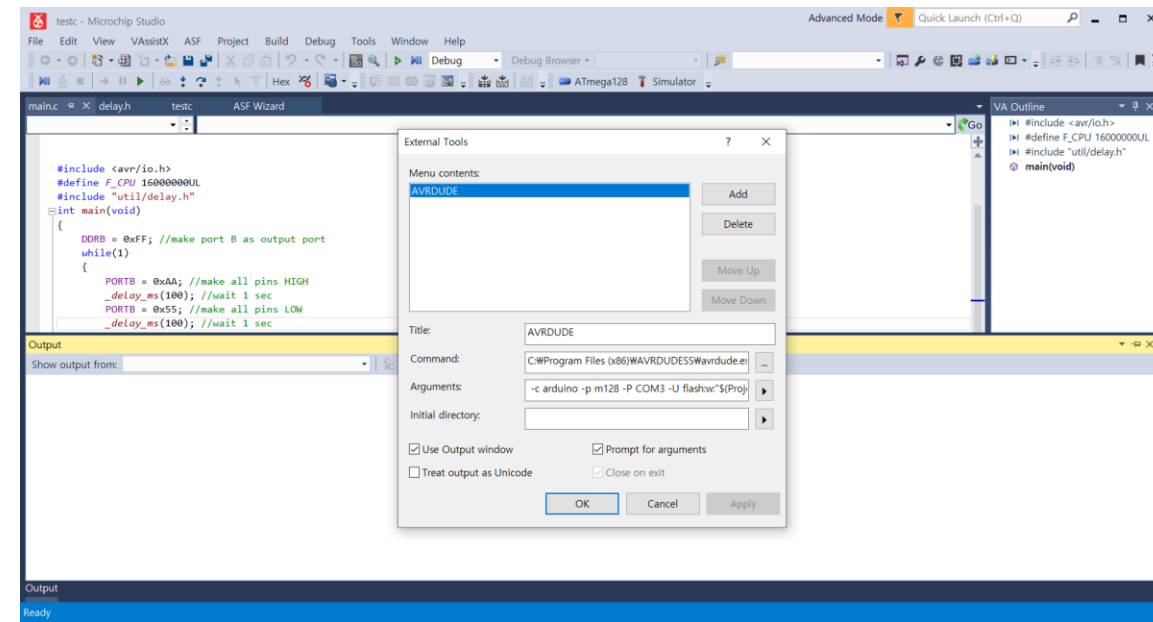
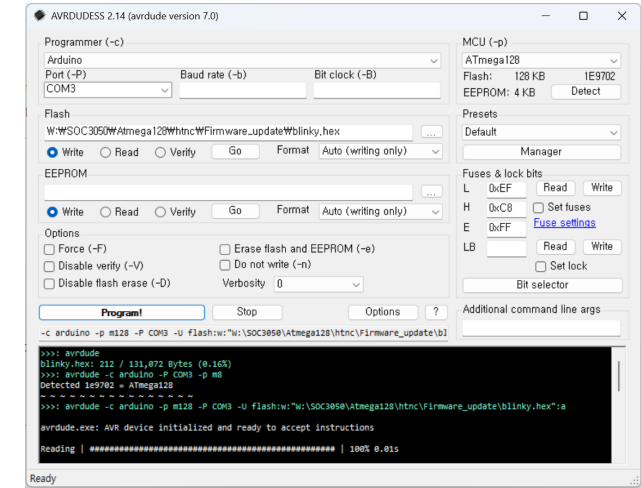
1. Install [AVRDUDESS](#)
2. Plugin Mini USB
3. Run [AVRDUDESS](#)
 - Detect
 - Programmer: Arduino
 - Port: COMx
 - Flash: blinky.hex
4. Go/Program!

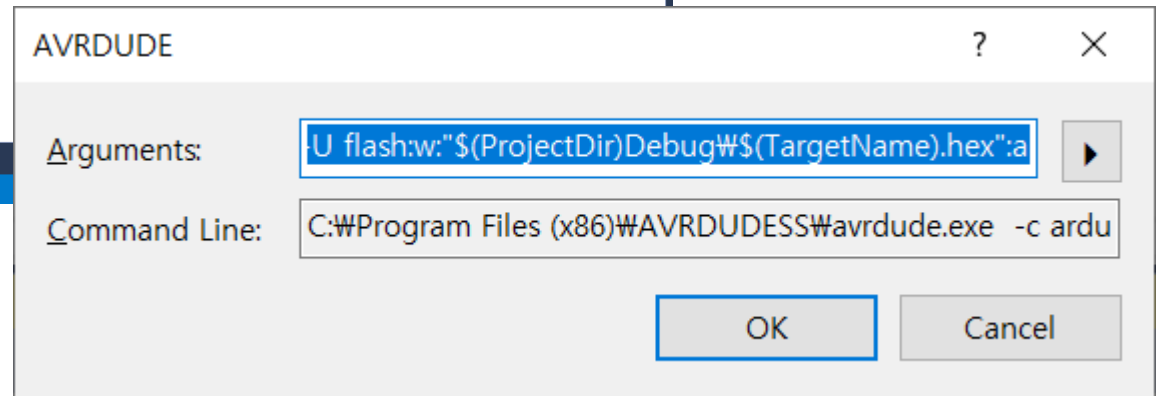
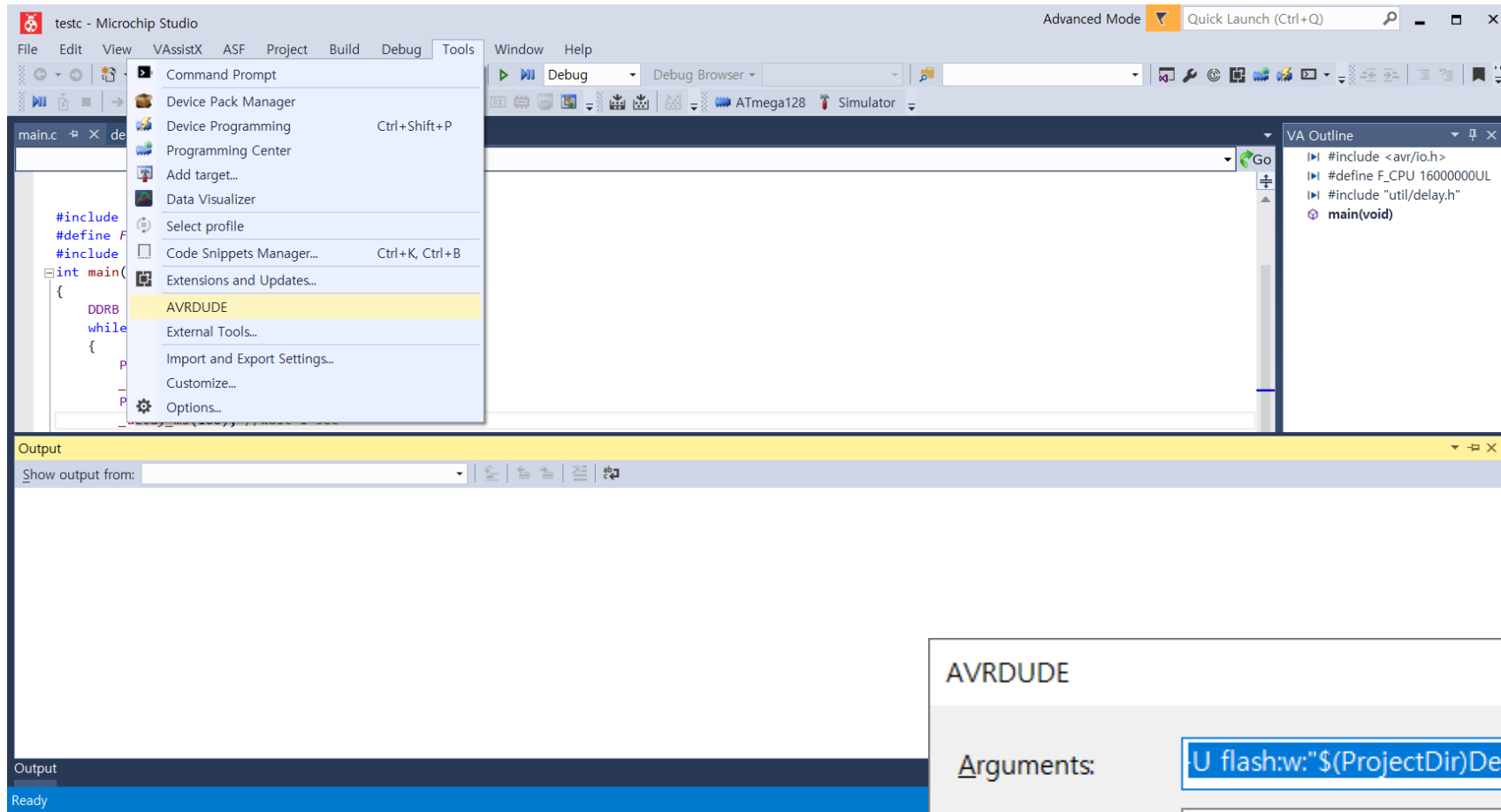


Microchip Studio Connection to AVRdude

- External Tools

- Title
- Command:
avrdudess/avrdude.exe
- Arguments:
 - c arduino -p m128 -P COM3 -U flash:w:"\$(ProjectDir)Debug\\$(TargetName).hex":a





MC Studio Stimuli: Debugging

```
1 // Example stimuli file, feeds any output on PINA back to PINB.
2 #10
3 DDRD = 0x00
4 PORTD = 0xFF //Pull-up
5 $repeat 1000
6     PIND = 0xFF
7     #10
8     PIND = 0x00
9     #10
10 $break
11 $endrep
```

```
-int main_interrupt (void)
{
    //apps
    DDRB = 0xFF;
    PORTB = 0xAA;

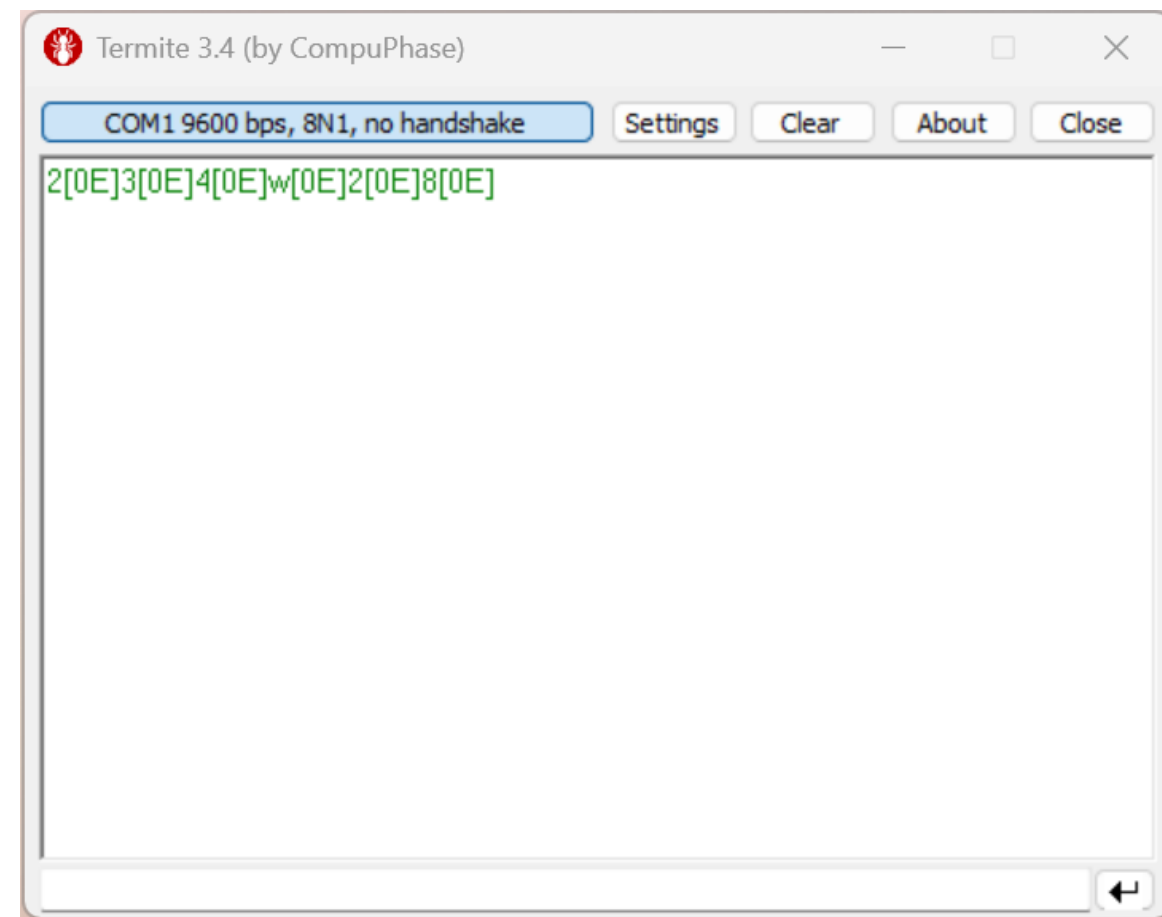
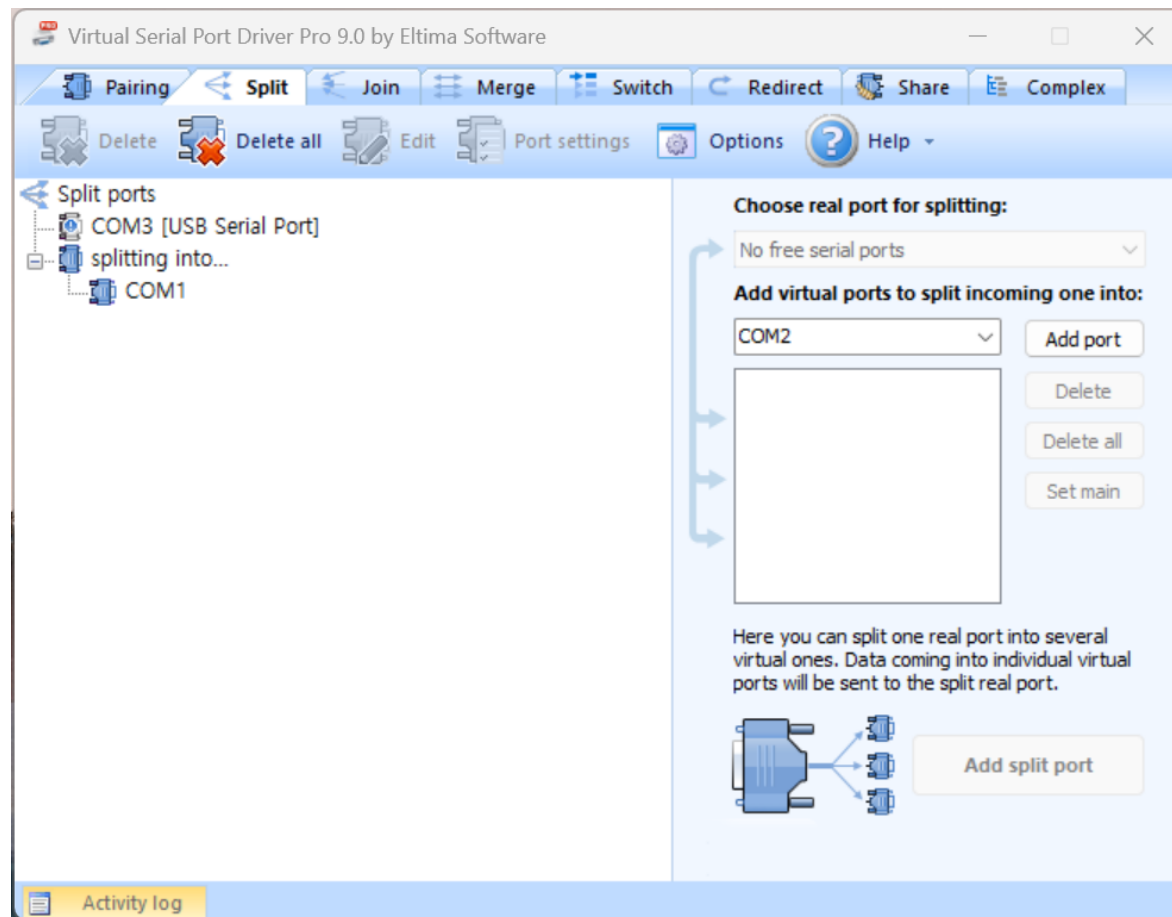
    //interrupt
    EICRA = 1<<ISC01;      //INT0 falling
    EICRB = 0x00;
    EIMSK = 1<<INT0;      //INT0 enabled
    sei();                 //Interrupt enabled
    //apps
    while (1) {}
    return 0;
}
```

```
-SIGNAL (INT0_vect){
    PORTB = ~PORTB;
}
```

- Preparation
 - c/asm file
 - Stim file
- MC Studio
 - Start Debugging
 - Execute Stimuli file
 - Click F5
- F5 stops where Stimuli file breaks.

Atmega128 + VSPD + Termite

- Serial communication between Termite and ATmega128 board bypassing MC Studio's Data Visualizer.
- Atmega128
 - Serial communication
 - COM3
- VSPD
 - Split ports
 - COM3 - COM1, COM2, ...
- Termite
 - COM1



Atmega128 board + MC Studio Data Visualizer

- Serial communication between Atmega128 board and Data Visualizer. No debugging mode.
- Download the code to Atmega128 board
- Run Data Visualizer

Main - Microchip Studio

Advanced ModeQuick Launch (Ctrl+Q)

FileEditViewVAssistXASFProjectBuildDebugvMicroToolsWindowHelp

Debugmain

HexATmega128Simulator

vMicro

main_adc.c_uart.cmain_serial.cMain.cData Visualizer

Configuration

Configuration Modules

External ConnectionVisualizationUtilitiesProtocols

Messages

11:45:52.595: EDBG Control Panel added.

Timestamp02:41:50.000044

Global scrolling

ThemeDark

DGI Control Panel

Serial Port Control Panel

USB Serial Port(COM3)

Baud rate9600ParityNoneStop bits1 bit

☒ DTR☐ RTS☒ Open Terminal☐ Autodetect protocols

Disconnect

Terminal 12

246

ClearAdd WrnHexadecimal ValuesShow TimestampAutomatically Scroll to End

Solution Explorer

Search Solution Explorer (Ctrl+.)

_init.c

_interrupt.c

_port.c

_timer2.c

_uart.c

Main.c

main_adc.c

main_graphics.c

main_inline.c

main_interrupt.c

main_interrupt.stim

main_joystick.c

main_memory.c

main_motors.c

main_port.c

main_serial.c

main_timer.c

ASF ExplorerSolution Explorer

Properties

Error List

Entire Solution0 Errors0 Warnings0 MessagesBuild + IntelliSense

Search Error List

Description

Project

File

Line

Output

Ready

SimulIDE + VSPD + Termite